Program	B. Tech.	Semester	I
Course	Advanced Engineering Mathematics I	Course Code	MATH 1059
Session	July-Dec 2025	Unit II	Integral Calculus

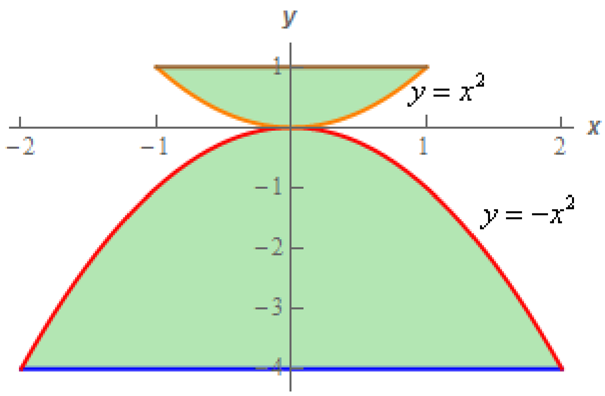
1. Find the value of the double integral

$$\int_{\mathcal{R}}\int |\, xy \,|\, dx\, dy$$

where the region $\mathcal{R}$ in 2 −space is enclosed by $x^2 + 4y^2 \geq 1$ and $x^2 + y^2 \leq 1$.

2. Change the order of the integration for $\int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{ax}} f(x,y)dy\, dx$.

3. Using suitable techniques of double integration, evaluate $\int_D \int (3 - 6xy)\, dx\, dy$, where D is the region shown below



4. Transform $\int_0^\infty \int_0^\infty e^{-(x^2+2xy\cos\alpha + y^2)} dx\, dy$ to polar coordinate system and hence show that its value is $\dfrac{\alpha}{2\sin\alpha}$ where $\alpha \in R - \{0\}$.

5. Evaluate $\iint xy\, dx\, dy$ by changing the variables over the region in the first quadrant bounded by the hyperbolas $x^2 - y^2 = a^2, x^2 - y^2 = b^2$ and the circles $x^2 + y^2 = c^2, x^2 + y^2 = d^2$ with $0 < a < b < c < d$.

Hint: Use suitable transformations $(x, y) \longrightarrow (u, v)$ to simplify!

6. Suppose a and b are real numbers such that $0 < a < b$; and the triple integral over the region bounded by the four planes $ax + b(y + z) = 1, x = 0, y = 0, z = 0$ is given by

$$\int_0^{\frac{1}{a}} \int_0^{p(x)} \int_0^{q(x,y)} dz\, dy\, dx$$

Prove that the function $p(x) - q(x, y)$ is independent of variable x.

7. Use triple integration to determine the volume of the region that is below $z = x - x^2 - y^2$ above $z = -\sqrt{(4x^2 + 4y^2)}$ and inside $x^2 + y^2 = 4$.

8. Evaluate the triple integral

$$\iiint_R (x^2 + y^2 + z^2) dx\, dy\, dz$$

Over the region bounded by $(x + y + z) = a, (a > 0), x = 0, y = 0, z = 0$.

9. Find the volume of the portion of the solid cylinder $x^2 + y^2 \leq 2$ bounded above by the surface $z = x^2 + y^2$ and bounded below by xy −plane.

10. Suppose $\mathcal{V}$ denotes the volume of the solid in $\mathbb{R}^3$ bounded by the following surfaces

$$x = -1, \qquad x = 1, \qquad y = -1, \qquad y = 1, \qquad z = 2, \qquad y^2 + z^2 = 2$$

Prove that $\mathcal{V} = (6 - \pi)$ cubic units.

11. Convert $\int_{-1}^{1} \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz\ dz\, dx\, dy$ into cylindrical coordinate system, hence compute the integral.

12. Prove the following results

a. $\int_{-1}^{1} (1 + x)^{m-1} (1 - x)^{n-1}\, dx = 2^{m+n-1} \beta(m, n)\ for\ m, n > 0.$

b. $\int_0^{\frac{\pi}{2}} \tan^n \theta\, d\theta = \frac{1}{2} \pi \sec\left(\frac{1}{2} n\pi\right).$

c. $\int_0^{\frac{\pi}{2}} (\sin x)^{\frac{8}{3}} (\sec x)^{\frac{1}{2}} dx = \frac{\Gamma\left(\frac{11}{6}\right)\Gamma\left(\frac{1}{4}\right)}{2\Gamma\left(\frac{25}{12}\right)}$.